## Conclusion

This project successfully demonstrates the design, simulation, and hardware implementation of a functional 8-bit CPU, providing a comprehensive understanding of fundamental computer architecture and digital system design. By integrating essential processor components including the Arithmetic Logic Unit (ALU), registers, program counter, control unit, instruction decoder, and memory the project illustrates the complete instruction execution cycle, from instruction fetch and decode to execution.

The combination of Logism for circuit simulation and the Basys 3 FPGA for hardware implementation, using Verilog HDL and Xilinx Vivado, enables real-time execution, testing, and debugging of the processor. This approach effectively bridges the gap between theoretical concepts and practical hardware realization while providing valuable insights into data flow, control signal generation, and processor operation.

The successful execution of sample programs, including multiplication and Fibonacci sequence generation, validates the correctness and reliability of the processor design. By emphasizing a modular architecture and a clear hardware-oriented implementation, the project offers an effective educational platform for exploring microprocessor fundamentals and digital logic design.

Overall, this work establishes a strong foundation for understanding processor architecture and hardware development. It serves as a valuable learning resource for students and researchers while providing a basis for future enhancements, such as expanding the instruction set, increasing processing capability, integrating peripheral interfaces, and exploring more advanced processor architectures.